

Differential Operators on a Graph

A theory guide for the graph engine

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Abstract

A mesh is a graph: cells are vertices, faces are edges. This note shows that once a problem lives on a graph, every differential operator a partial differential equation needs – first derivative, second derivative, any order, curl, and each operator’s adjoint – is built from three elementary steps, each of which is one walk over the edges. A solver for a physical problem then writes no loops of its own. It chooses operators, hands them coefficients, and the graph engine does the rest. Every claim made here is proved by arithmetic short enough to check by hand, and each construction is drawn before it is written.

1 The one-line law, and where this note fits

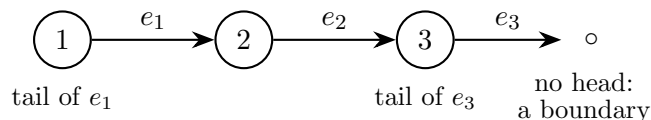
The engine rests on one sentence: split a graph into parts, work on the parts, put the answer back together.

$$G' = P^{-1}(A(P(G)))$$

P is the partitioner, P^{-1} the assembler, and A an operation. This note is about the operations A : the ones that turn a field of values into its derivatives. When these exist, a solver for a physical problem – heat spreading through a plate, a quantity carried along a flow – is nothing but a *choice* of operators and coefficients. The physics chooses; the graph computes. That division of labour is the point of the whole design, and section 10 spells it out.

2 Graphs and fields

A graph is vertices joined by edges. Every edge has a tail and a head, so every edge points somewhere. An edge may also have a tail and *no* head: it starts at a vertex and stops. That is how a boundary is written – a face on the outside of a mesh touches one cell only, and no imaginary cell is invented beyond it.

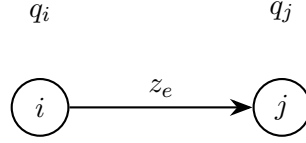


A *vertex field* is a value per vertex; call the values q_v . An *edge field* is a value per edge; call the values z_e . Nothing more is needed. (A reader who knows the exterior-calculus literature may recognise these as 0- and 1-cochains; the names are never needed again.)

Each edge also carries a length h_e , the spacing between its two ends, and each vertex carries a measure m_v – on a mesh, the volume of its cell. On the uniform chains used for the proofs below, $h_e = h$ and $m_v = h$ everywhere.

3 Three elementary steps

Everything in this note is built from three ways of moving values between vertices and edges. Two go from vertices to edges; one comes back.



The average step S puts on each edge the average of its two end values, or – when a direction matters – the value at one end, chosen by the sign of the coefficient the operator carries:

$$(Sq)_e = \frac{1}{2}(q_i + q_j) \quad \text{or} \quad (Sq)_e = q_i \quad \text{or} \quad q_j.$$

The difference step G puts on each edge the difference of its two end values over the spacing:

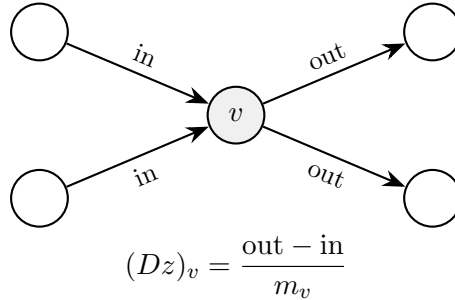
$$(Gq)_e = \frac{q_j - q_i}{h_e}.$$

This is the slope of q along the edge.

The fold step D brings edge values back to the vertices. Each vertex adds up the values on edges leaving it, minus the values on edges entering it, and divides by its measure:

$$(Dz)_v = \frac{1}{m_v} \left(\sum_{\text{edges leaving } v} z_e - \sum_{\text{edges entering } v} z_e \right).$$

An edge with no head contributes to its tail alone.

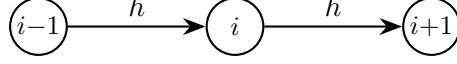


At a boundary the missing end needs a value. The operator stores one (a single number, or one per boundary edge), and S and G use it in place of the value that is not there. This is where a boundary condition's *data* enters; the operator itself does not change.

One convention to fix now, because two are in use. The fold above is *out minus in*; it makes the derivative formulas of the next section come out with their textbook signs. The balance operation in the engine folds the other way – *in minus out* – because a balance counts what a vertex *gains*. The two folds differ by a sign and both are stated where they appear. Nothing deeper is going on.

4 Any order, by repeating the steps

Apply the steps in alternation and the derivatives appear in sequence. On the chain below, with uniform spacing h and measure $m = h$:



order	recipe	result at an interior vertex	name
0	$c q$	$c q_i$	the term itself
1	$D S q$	$c \frac{q_i - q_{i-1}}{h}$	first derivative
2	$D G q$	$c \frac{q_{i+1} - 2q_i + q_{i-1}}{h^2}$	second derivative
3	$D G D S q$	(the stencil widens)	third
4	$D G D G q$	fourth difference over h^4	fourth
n	one more step	reaches n rings of neighbours	n -th

The recipes on the edge side read the same way: order 0 is $S q$, order 1 is $G q$, and order n applies G to the vertex result of order $n - 1$. The coefficient c is applied once, at the outermost step. (The order-2 recipe is what the literature calls the graph Laplacian; the name is not needed to use it.)

Each step consults only an edge's two ends, so each step widens the reach of the answer by exactly one ring of neighbours: the order- n operator sees precisely the vertices a walk of n edges can reach. Arbitrary order is therefore one integer and a loop – nothing about a graph makes order 7 harder than order 2.

Claim 1 (The second derivative is exact on a parabola). *Take the uniform chain with $q_v = v^2$ and $c = 1$. The whole computation fits in three lines. The difference step gives, on the two edges at i ,*

$$(Gq)_{(i-1,i)} = \frac{i^2 - (i-1)^2}{h} = \frac{2i-1}{h}, \quad (Gq)_{(i,i+1)} = \frac{(i+1)^2 - i^2}{h} = \frac{2i+1}{h},$$

and the fold step subtracts them and divides by $m = h$:

$$(DGq)_i = \frac{1}{h} \left(\frac{2i+1}{h} - \frac{2i-1}{h} \right) = \frac{2}{h^2}.$$

In the chain's coordinates $x = vh$ this is the statement that the second derivative of x^2 is 2, exactly. On a straight line $q_v = av + b$ the same two lines give zero. $\square$

Claim 2 (Order four is exact on x^4). *With $q_v = v^4$ and $h = 1$, one application of DG gives, by the same arithmetic,*

$$(DGq)_i = (i+1)^4 - 2i^4 + (i-1)^4 = 12i^2 + 2,$$

and applying DG once more to $12i^2 + 2$ gives, by the parabola claim, $12 \cdot 2 = 24$: the fourth derivative of x^4 , exactly, at every vertex two rings inside the boundary. $\square$

Odd orders above one mix a one-sided step with centred ones, so they match a hand-computed stencil rather than a single calculus formula; the test suite checks them against that stencil.

5 Coefficients that vary: the non-uniform case

Nothing above requires the coefficient, the spacing, or the boundary value to be one number. Give every edge its own:

$$z_e = c_e \frac{q_j - q_i}{h_e}, \quad (Dz)_v = \frac{1}{m_v} (\text{out} - \text{in}),$$

and order 2 becomes the operator $\text{div}(k \text{ grad } q)$ with a conductivity that varies in space – each edge carries its own c_e . The uniform case is the special case of equal values.

This is also precisely the finite-volume method, which is why this machinery replaces the hand-written assembler. Read the dictionary:

mesh quantity	graph home	used by
cell	vertex	everything
interior face	edge with a head	S, G, D
boundary face	edge, no head	boundary values
cell volume V_v	vertex measure m_v	the fold step
face area A_e and conductivity k_e	edge coefficient $c_e = k_e A_e$	the difference step
centre-to-centre distance d_e	edge spacing h_e	the difference step

With those entries, the order-2 vertex operator at cell v reads

$$\frac{1}{V_v} \sum_{\text{faces of } v} \pm k_e A_e \frac{q_{\text{other}} - q_v}{d_e},$$

which is the finite-volume discretisation of $\text{div}(k \text{ grad } q)$, term for term. The engine never needs to know it is doing finite volume; the mesh hands the graph its geometry, and the formula falls out.

6 From operators to equations

A transport equation,

$$\frac{\partial q}{\partial t} = -\text{div}(u q) + \text{div}(k \text{ grad } q) + s,$$

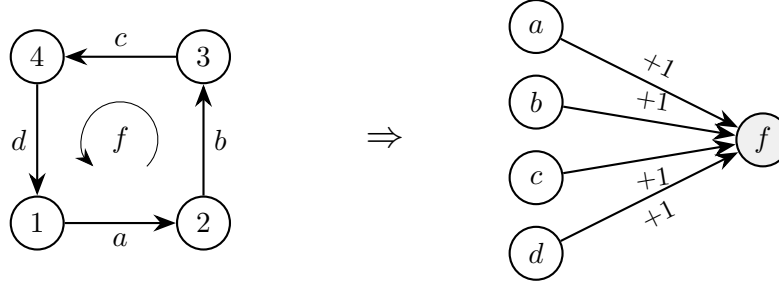
asks for exactly three configured operators and a source: an order-1 vertex operator with coefficient u (the sign of u choosing the sampled end – values are taken from the side the flow comes from), an order-2 vertex operator with coefficient k , and an order-0 term for s . The engine’s balance operation sums such terms and a time integrator marches the result. Boundary conditions are not special operators: they are the stored values at the headless edges, per-edge when the boundaries differ.

Signs deserve one honest sentence. Physical laws carry their own sign conventions – a flux runs *down* a gradient, so a model writes its diffusion coefficient with the sign its law requires. The graph layer computes derivatives; which sign a term enters an equation with is the model’s statement, made in the model’s own class.

7 Going around instead of along: curl

The derivative along edges is not the only derivative. Around a loop of edges there is another one: add up the edge values as you walk the loop. If the values came from differences of a vertex field, the walk telescopes to nothing; if the values circulate, it does not. That sum is the curl.

Where does it live? A loop of edges bounds a face, and a face is not a vertex or an edge – but *which edge borders which face* is itself a graph. Build it: one vertex for every edge of the original graph, one vertex for every face, and a connection wherever an edge borders a face, carrying $+1$ if the edge agrees with the direction of the walk around the face and -1 if it opposes it.



On this second graph, the edge values of the original become a vertex field, and the fold step – the same D as everywhere in this note, with the ± 1 as its per-edge coefficients – collects the walk around each face. The curl is the fold step on the border graph. Nothing new is built.

Claim 3 (The curl of a difference field is zero). *Walk the square: the edge values of a difference field are $z_a = q_2 - q_1$, $z_b = q_3 - q_2$, $z_c = q_4 - q_3$, $z_d = q_1 - q_4$. Their sum,*

$$(q_2 - q_1) + (q_3 - q_2) + (q_4 - q_3) + (q_1 - q_4) = 0,$$

cancels term against term, for any values q . □

A field that circulates – say $z_a = z_b = z_c = z_d = 1$ – sums to 4, not zero, so the construction detects rotation, which is its job. The same move repeats upward: loops of faces bound volumes, and their border relation is again a graph, for whoever needs it.

8 Adjoints: the same walk, backwards

Every operator here is a walk over the edges. Its adjoint – the operator A^* with $\sum_v (Aq)_v p_v = \sum_v q_v (A^*p)_v$, the thing an adjoint (sensitivity) solver consumes – is the same walk with every edge reversed.



To see it, take the plain sums ($h = m = 1$ keeps the arithmetic bare; weights ride along unchanged). Pair a vertex field p with a fold:

$$\sum_v p_v (Dz)_v = \sum_e z_e (p_{\text{tail}(e)} - p_{\text{head}(e)}) = - \sum_e z_e (Gp)_e.$$

The first equality is only a change of viewpoint: instead of walking vertices and asking which edges touch each one, walk edges and ask which vertices each one touches – every edge hands $+z_e$ to its tail’s row and $-z_e$ to its head’s row. So

$$D^* = -G, \quad G^* = -D,$$

and the one-sided average step’s adjoint folds, unsigned, onto the end it sampled. Reversing every edge swaps tail with head, which swaps out with in, which is exactly these transposes: the reverse walk.

Adjoints of compositions run the transposed steps in reverse order. Two consequences follow immediately:

$$(DG)^* = G^* D^* = (-D)(-G) = DG,$$

so the second-derivative operator is its own adjoint whenever the same per-edge coefficients sit on both steps – and the order-1 operator’s adjoint samples from the *other* end and folds the other way, which is precisely the reversed-flow operator a sensitivity solve requires. The engine gets both by a flag: run the walk backwards. No second implementation exists to drift out of agreement with the first.

9 Cut into parts, the operators do not notice

The engine’s three laws state that partitioning is invisible to results:

$$P^{-1}(P(G)) = G, \quad P^{-1}(P(G, D)) = (G, D), \quad A(G, D) = P^{-1}(A(P(G, D))).$$

The steps of this note obey the third law for the same reason the balance does: each step reads an edge’s two ends, and a part carries – as its borrowed vertices – every end its owned edges need. A part computes its owned share exactly; assembly collects owned values only; the pieces sum to the whole.

One reduction subtlety travels with partitioning and is worth restating. A quantity like an average must be folded across parts *unfinished*: two parts averaging 2 and 7 combine to 4 – twenty over five – not 4.5. The engine’s reductions carry their running sums and counts together and divide once, at the end.

10 What a physics class actually does

Everything above belongs to the engine. A model of a physical problem now delegates:

1. **choose operators** – which orders appear in its equation;
2. **fill coefficients** – from geometry ($c_e = k_e A_e$, $h_e = d_e$, $m_v = V_v$) and, for a nonlinear law, from the current state, refreshed each evaluation;
3. **state its signs** – its own flux convention, in its own class;
4. **set boundary values** – per headless edge, from its boundary conditions.

Worked example, heat in a plate: $\partial q/\partial t = \text{div}(k \text{ grad } q)$ with insulated boundaries. The model builds one order-2 vertex operator; fills $c_e = k_e A_e$, $h_e = d_e$, $m_v = V_v$ from the mesh; sets each boundary edge's stored value equal to its cell's value (no difference across the boundary, so no flow); and hands the operator to the balance and the balance to a time integrator that already exists. The model contains no loop over cells or faces. That absence is the design goal.

Three pieces complete the full replacement of the old hand-written assembler, all already mapped: the mesh migration that feeds the coefficient arrays from stored geometry; state-dependent coefficients recomputed per evaluation (the delegation point for nonlinear laws); and systems of equations, which ride on the fields' `num_components` unchanged.

11 The dictionary

in this note	in the engine	kind
vertex field q	<code>vertex_field</code>	type
edge field z	<code>edge_field</code>	type
average step S	<code>average_step</code>	kernel
difference step G	<code>difference_step</code>	kernel
fold step D	<code>fold_step</code>	kernel
operator, edge side	<code>edge_differential_operator</code>	type
operator, vertex side	<code>vertex_differential_operator</code>	type
order n	<code>order</code>	component
c , c_e	<code>coefficient, coefficients(:)</code>	component
h , h_e	<code>spacing, spacings(:)</code>	component
m_v	<code>measures(:)</code> on the vertex side	component
boundary value	<code>boundary_value, boundary_values(:)</code>	component
the reverse walk	<code>adjoint flag</code>	component
order 1, edge side	<code>gradient(...)</code>	constructor
order 0, edge side	<code>interpolation(...)</code>	constructor
order 1, vertex side	<code>divergence(...)</code>	constructor
order 2, vertex side	<code>laplacian(...)</code>	constructor
sum of terms	<code>balance</code>	type

Every claim in this note reappears in `test/graph-contract` as a check with the same numbers: the parabola's 2, the fourth power's 24, the average's 4-not-4.5, the square loop's 0 – and its nonzero companion for a circulating field, because a check that cannot fail proves nothing.